

Nomodynamics: The Mathematics of Self-Amending Law

A founding note

The Treeline Program

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Abstract

Game theory formalized games with fixed rules; Conway’s deletion of the *players* gave cellular automata and the Game of Life. The orthogonal deletion—removing the *fixity of the rules*—was never taken, although rules that amend rules are humanity’s oldest formal practice. We introduce a minimal, parameter-free mathematical object native to that vacancy: a system whose only substance is laws acting on laws. Its founding (“own-kind”) semantics turns out to be exactly solvable, with short theorems that carry immediate institutional readings: fully occupied codes are frozen (*gridlock*); stable codes are always fully blocked (*dead letters*); and on the integer line *the eldest law cannot be repealed*, which makes free-moving law-packets impossible—while on circular codes, which have no eldest law, moving packets provably exist. The solvable sector exhibits a small charismatic fauna (colonizers, sunset clauses, welds, refraction, rotating packets, and an aperiodic avalanche machine locked to powers of two), and its unique escape hatch—laws amending *other* laws—is located by theorem.

1 The vacancy

Three observations, from a systematic survey of how new mathematical ontologies have historically been found, converge on one empty cell. (i) The deletion grid: von Neumann–Morgenstern formalized 2-player fixed-rule games; puzzles are the 1-player case; Conway’s “0-player games” are cellular automata. Along the orthogonal axis—rule fixity—no elementary mathematics exists. (ii) The reduction shadow: computability theory’s founding reduction, “without loss of generality the program is fixed,” is valid for *what can be computed* (universality lets a fixed machine simulate a self-modifying one) but silently discards the *behavioral* questions: what do small self-amending systems actually do? Nobody looked, because the reduction said one need not. (iii) The practiced verb: constitutions and statutes—self-amending by essence—are the oldest running formal systems, yet “amend” was never objectified the way Church and Turing objectified “compute.” The fragments that exist (von Neumann’s rule-carrying constructor cells; Suber’s game *Nomic*, 1982; self-modifying Petri nets, 1978; algorithmic chemistry) never produced a minimal canonical object, a census, or a theorem.

2 The object

Definition 1 (nomic chain). A *law* is a triple $(a, b, c) \in \{-1, 0, 1\}^3$ placed at a position $i \in \mathbb{Z}$. A *code* is a finite set S of placed laws. A placed law $(i, (a, b, c)) \in S$ is *active* if some law stands at $i + a$ and no law stands at $i + b$. Time is synchronous: every active law flips the presence of *its own kind* at $i + c$, flips resolving by parity. Nomic *rings* are the same system on $\mathbb{Z}/m$; the d -dimensional and window- w versions replace $\{-1, 0, 1\}$ by offset sets in $\mathbb{Z}^d$.

There is no board and no external rule table: the law is the only substance. The system is parameter-free. All claims below were pre-registered as predictions or risks before the first simulation, then settled by censuses with machine-checked certificates (~ 15.3 million certified runs for the no-go results; complete enumerations where stated).

3 First theorems: the solvable sector

Theorem 2 (Gridlock). *In a fully occupied region every law’s vacancy guard fails; a solid code is frozen. All dynamics is surface dynamics.*

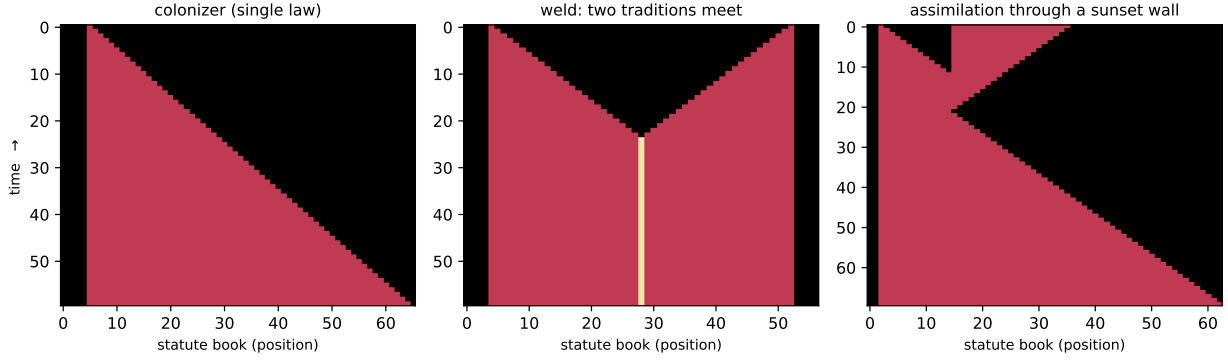


Figure 1: Window-1 nomic chains (time downward). Left: the *colonizer* $(0, 1, 1)$ —“while I stand and my right is vacant, enact my kind to the right”—only its frontmost copy is ever active. Middle: two traditions meet and *weld* into frozen code with a double-law seam. Right: a colonizer meets a self-eroding *sunset wall*. The wall repeals itself from its far end while the front stands blocked; the front then fills the vacated ground, so a wall of length L is cleared at time exactly $\max(2L, x_0 + L)$ — the “refraction index 2” is those two speed-1 processes in sequence. The colonizer never converts the wall: it waits for the old code to sunset and takes the vacancy.

Lemma 3 (Single Author). *The only law that can ever flip kind $k = (a, b, c)$ at position j is the kind- k law at $j - c$. Hence parity and idempotent (“OR”) resolution coincide identically, in every dimension, and per-kind dynamics is occupancy-modulated linear algebra over $\mathbb{F}_2$.*

Theorem 4 (Dead Letter). *A code is a fixed point iff every law in it is blocked. In particular balanced constitutions—codes in perpetual self-cancelling activity—do not exist: stability is always gridlock.*

Theorem 5 (Anchor). *On $\mathbb{Z}$ (any window, any dimension, any guard predicate, either resolution semantics), own-kind toggles of kind t land only at t ’s own offset c_t ; consequently the extremal law on the trailing side of a finite code is never targeted: the eldest law cannot be repealed. No finite code satisfies $\Phi^p(S) = \sigma^d(S)$ with $d \neq 0$: free gliders are impossible.*

Corollary 6 (Ring rotors). *On $\mathbb{Z}/m$ there is no eldest law, and the obstruction genuinely vanishes: the minimal moving law-packet of nomodynamics is three laws of the single kind $(0, 1, -1)$ at cells $\{1, 2, 5\}$ of $\mathbb{Z}/6$, hopping $m/2 = 3$ cells per step (Fig. 2); such rotors exist on every even $m \geq 6$ and on no odd ring (≤ 3 -law sweeps complete for $m \leq 7$). Entrenchment is a theorem of linear order: linear statute books are anchored by their oldest provision; circular codes can revolve.*

Remark 7. The lemma explains the observed clockwork: every exact cycle found across $\sim 140,000$ seeds has period a power of two, and single-kind ring codes (the “Sunset Parliament”) attain exact maximal periods 15, 63, 341 at resonant circumferences $m \equiv 2 \pmod{4}$ —orders of $\mathbb{F}_2$ -polynomials governing constitutional cycles.

4 Fauna of the solvable sector

The one-dimensional chain is a calculus of fronts (Fig. 1): *colonizers* (2 of the 27 kinds) fill the line at speed 1; the *sunset clause* $(0, -1, 1)$ forever enacts a subordinate its own presence then blocks (period 2); *sunset codes* dissolve from their newest end; colliding fronts *weld*; porous codes support one-way *conversion waves* (speed $2/3$); walls facing a front sort into opaque ($19/27$), pulsing ($3/27$), and effectively transparent ($5/27$) classes. In two dimensions a *ray-confinement theorem* pins each kind to one axis ray (quadrants never fill), and the fauna adds 239 certified half-turn *pinwheel rotors*, Pascal-column growers with $|S_t| = 2^{\text{popcount}(t)}$, and the Jubilee Code (Fig. 2, right). Finite rings admit an exact fixed-point count (transfer matrix, brute-force verified), a predecessor algebra with Garden-of-Eden fraction peaking $\approx 33\%$ at mid-density, and a unique minimal taut architecture (the two-law *mutual-veto* pair).

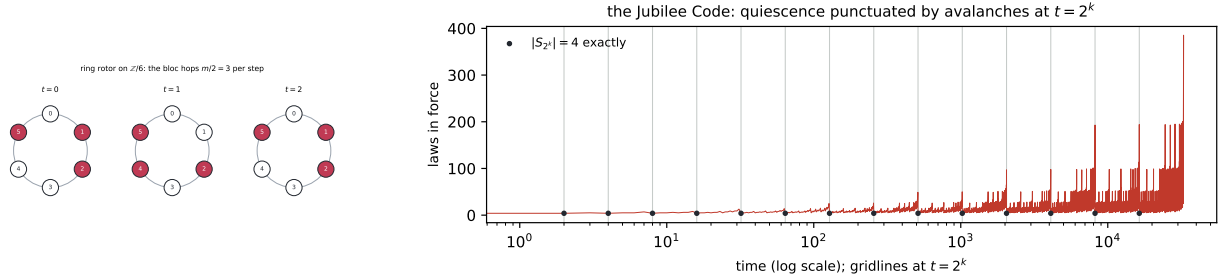


Figure 2: Left (2): the minimal ring rotor. Right: the *Jubilee Code*, a ~ 26 -law two-dimensional machine, aperiodic through 300,000 fully-hashed steps and quiescent except at $t = 2^k - 1$, when the counter fills and the whole code ignites; at $t = 2^k$ itself exactly four laws stand, every time (16/16, complete through $t < 2^{17}$), and the crest height doubles every second power of two. A binary counter native to law-space, and an attractor: 791 of 60,000 random seeds converge to its family.

5 The frontier, located by theorem

Everything above holds because laws amend only their own kind—the tame simplification. Real law amends *other* law, and by the Single Author Lemma, *cross-amendment* (effects that target other kinds) is precisely the escape from the solvable sector: multi-author targets break both the anchor and the linearity. First probes confirm the door is real: parity and OR semantics genuinely diverge there, and diagonal motion—impossible under ray confinement—unlocks in 2D. Open problems: (1) does cross-amendment admit free gliders on $\mathbb{Z}$, or does a deeper no-go extend? (2) is any small cross-amendment universe computation-universal? (3) classify ring-rotor families (which m , which hop groups); (4) the Jubilee attractor’s basin: why do $> 1\%$ of random seeds fall in? (5) the reception question the founding practice suggests: which theorems of own-kind nomodynamics survive into semantics faithful to actual amendment procedures (supersession, entrenchment clauses, quorum guards)?

Methods. Definition and predictions were frozen before the first run; censuses carry machine-checked certificates (recurrence, translation-recurrence, rotation-recurrence); complete enumerations: 1- and 2-law seeds ($W=1$), ≤ 4 -law/5-cell patterns (8.1M), ring sweeps $m \leq 12$; the Anchor and Dead Letter theorems have short combinatorial proofs checked against 15.3M runs with zero violations. Full records: the Treeline program archive, [phase6/](#).